

Mental Health and Psychosocial Outcomes Among Individuals With Mayer-Rokitansky-Küster-Hauser (MRKH) Syndrome: A Scoping Review

Running title: Mental health in MRKH syndrome: a scoping review

Ifeoluwanimi P. Shobayo, BSc, MSPHc¹; Paul Okojie, PhD¹; Robyn Anderson, PhD¹

¹Department of Public and Community Health, Liberty University, Lynchburg, Virginia, USA

Corresponding author: Ifeoluwanimi P. Shobayo, Department of Public and Community Health, Liberty University, Lynchburg, VA, USA. Email: [\[corresponding author email\]](#)

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Abstract

Background: Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome is a rare congenital condition, affecting approximately 1 in 4,500–5,000 female births, characterized by uterovaginal agenesis in individuals with a 46,XX karyotype and functional ovaries. Typically diagnosed during adolescence, MRKH is associated with significant psychosocial burden, yet mental health outcomes and coping mechanisms remain insufficiently synthesized, limiting their integration into clinical care and public health planning. This scoping review mapped the mental health outcomes, coping mechanisms, and healthcare system gaps reported among individuals with MRKH.

Methods: Following Arksey and O'Malley's framework and the PRISMA Extension for Scoping Reviews (PRISMA-ScR), we searched PubMed/MEDLINE, Scopus, PsycINFO, and CINAHL for studies published between January 2019 and March 2026. Eligible studies were primary quantitative, qualitative, and mixed-methods research reporting psychological outcomes and/or coping mechanisms in MRKH populations.

Results: Of 516 records identified (361 after deduplication), 34 studies met inclusion criteria. Depression and anxiety were the most frequently reported outcomes ($n = 26$; 76%), followed by reduced quality of life and body image concerns ($n = 22$; 65%), psychosexual and relational challenges ($n = 19$; 56%), and broader psychological distress ($n = 17$; 50%). Coping mechanisms were documented in 21 studies and healthcare system gaps in 18. Most studies originated from Europe ($n = 18$) and North America ($n = 8$).

Conclusions: MRKH-related psychosocial burden is substantial yet under-integrated into care models. Multidisciplinary, mental health-inclusive care and greater research investment, particularly in underrepresented and low-resource settings, are urgently needed.

Introduction

Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome, also termed Müllerian agenesis or Müllerian aplasia, is a congenital condition characterized by absence of the uterus and upper two-thirds of the vagina in individuals with a 46,XX karyotype, functional ovaries, and typical secondary sexual development. It affects approximately 1 in 4,500–5,000 female births;¹ a nationwide Danish registry study estimated a prevalence of 1 in 4,982 live female births.² MRKH is classified as type I (isolated uterovaginal aplasia) or type II (accompanied by extragenital anomalies, most often renal, skeletal, auditory, or cardiac).³ Diagnosis is usually made in adolescence during evaluation for primary amenorrhea,^{1,3} a developmental period that is critical for identity formation, body image, and social relationships.

The diagnosis carries profound implications for fertility, psychosexual development, and self-image. Management commonly includes vaginal dilation as first-line treatment, surgical creation of a neovagina when indicated, and fertility options such as gestational surrogacy and, more recently, uterus transplantation.^{1,3} The American College of Obstetricians and Gynecologists identifies psychosocial counseling as one of the most important components of effective management.¹ Earlier studies documented elevated psychological distress and lower self-esteem among women with MRKH compared with controls,⁴ higher anxiety scores among affected adolescents,⁵ and poorer mental health–related quality of life, higher anxiety, and impaired sexual wellness relative to normative data.⁶ A review of this early literature concluded that creating a neovagina alone does not ensure a good psychological outcome and that psychological support at critical times can be helpful,⁷ while qualitative work described how young women manage intimacy, a sensitivity to difference, and threats to identity after diagnosis.⁸

Two systematic reviews have since synthesized the psychological, quality-of-life, and sexual outcomes of MRKH.^{9,10} However, to our knowledge, the more recent literature has not been mapped with a public health lens that jointly considers mental health outcomes, the coping mechanisms individuals use, and the healthcare system gaps that shape access to psychosocial support. As a result, psychosocial outcomes remain insufficiently integrated into clinical and public health frameworks for this population.

A scoping review is well suited to mapping a heterogeneous body of quantitative, qualitative, and mixed-methods evidence and identifying gaps for future research and practice.^{11,12} This scoping review therefore addressed the following research question: *What mental health outcomes and coping mechanisms are reported among individuals with MRKH syndrome, and what healthcare system gaps exist?*

Methods

Study Design

This scoping review followed Arksey and O'Malley's five-stage methodological framework¹¹—(1) identifying the research question; (2) identifying relevant studies; (3) selecting studies; (4) charting the data; and (5) collating, summarizing, and reporting the results—and is reported in accordance with the PRISMA Extension for Scoping Reviews (PRISMA-ScR; Appendix C).¹² [Protocol registration: state the registry and ID (e.g., OSF) or “A review protocol was not registered.”]

Eligibility Criteria

Eligibility was defined using the Population–Concept–Context approach. The *population* was individuals diagnosed with MRKH syndrome; the *concepts* were psychological or mental

health outcomes and coping mechanisms; and the *context* was any healthcare or community setting in any country. We included primary quantitative, qualitative, and mixed-methods studies published between January 2019 and March 2026 that reported psychological outcomes and/or coping mechanisms among individuals with MRKH. Studies focused solely on anatomical, surgical, or fertility outcomes were excluded. [Confirm any additional criteria applied, e.g., language (English only?), peer-reviewed publications only, and exclusion of reviews, case reports, editorials, and conference abstracts.]

Information Sources and Search Strategy

Four electronic databases were searched: PubMed/MEDLINE, Scopus, PsycINFO, and CINAHL, limited to January 2019 through March 2026. Search strategies combined controlled vocabulary and free-text terms for MRKH (e.g., “Mayer-Rokitansky-Küster-Hauser,” “MRKH,” “Müllerian agenesis,” “vaginal agenesis”) with terms for mental health and coping (e.g., “depression,” “anxiety,” “psychological distress,” “quality of life,” “body image,” “coping”). The full search strategy for each database is provided in Appendix A. [Date the final search was run: _____. Verify that the example terms above match the search you actually ran.]

Study Selection

Records were exported to [reference manager/screening software, e.g., Zotero, Covidence, Rayyan], and 155 duplicates were removed. The remaining 361 records were screened against the eligibility criteria [by two independent reviewers; disagreements resolved through discussion or by a third reviewer — edit to reflect your actual process]. [Report the number of full-text articles assessed for eligibility and the reasons for full-text exclusion; PRISMA-ScR item 14 requires this.]

Data Charting

Data were charted using a standardized form capturing author, year, country, study design, sample characteristics, outcome measures, reported mental health outcomes, coping mechanisms, and healthcare system gaps. [Confirm charted items and whether charting was piloted or completed in duplicate.]

Synthesis

Findings were collated using a descriptive numerical summary—the number and proportion of included studies reporting each outcome domain—and a qualitative content analysis that grouped findings into mental health outcome domains, coping mechanisms, and healthcare system gaps. Because a single study could report multiple domains, domain counts are not mutually exclusive. Consistent with scoping review methodology, a formal critical appraisal of study quality was not performed.^{11,12} [Confirm; if you did appraise quality, describe the tool used.] Ethical approval was not required because this review analyzed published data.

Results

Selection of Sources of Evidence

The database searches identified 516 records. After removal of 155 duplicates, 361 records were screened, of which 327 were excluded for not meeting inclusion criteria. A total of 34 studies were included in the review (Figure 1).

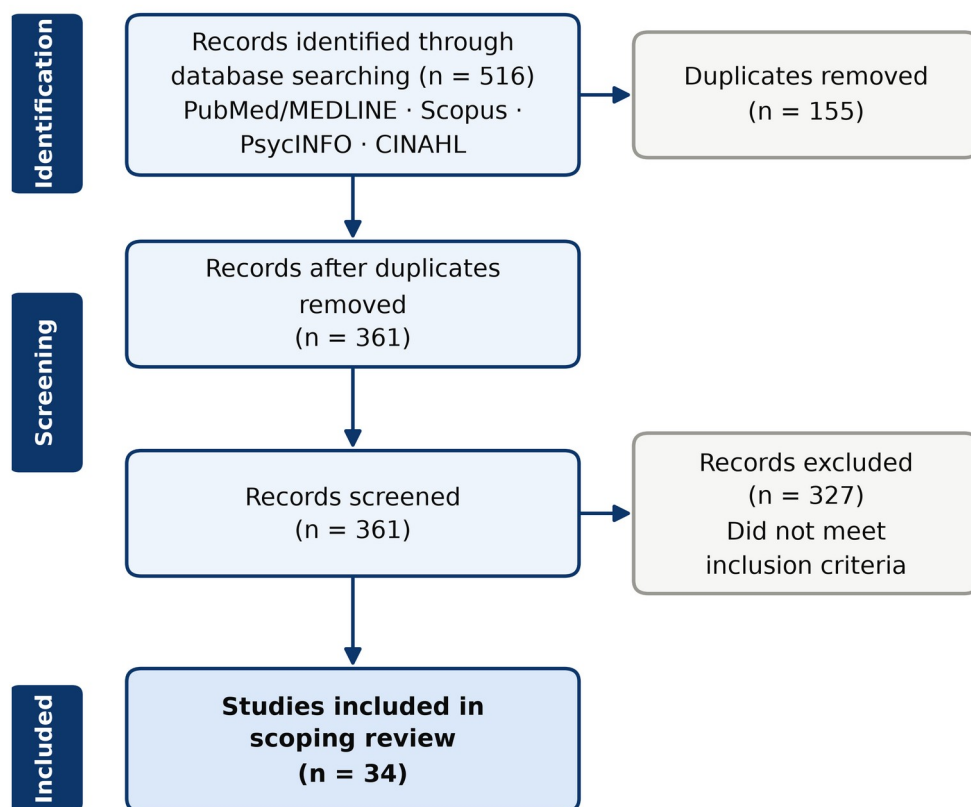


Figure 1. PRISMA-ScR flow diagram of study selection. [Add a full-text assessment box (number assessed and number excluded with reasons) if full-text screening was conducted as a separate stage.]

Characteristics of Included Studies

The 34 included studies comprised 16 quantitative studies (47.1%), 11 qualitative studies (32.4%), and 7 mixed-methods studies (20.6%); 11 of the quantitative studies used a cross-sectional design (Table 1; Figure 2). Studies were concentrated in high-income regions: 18 (52.9%) originated in Europe and 8 (23.5%) in North America, together accounting for 26 of 34 studies (76%). Five studies (14.7%) were conducted in Asia, 2 (5.9%) in Africa, and 1 (2.9%) in South America; no included study originated in Oceania (Figure 3). Characteristics of each included study are summarized in Appendix B. [Add the range of sample sizes, participant ages, and publication years from your extraction table.]

Table 1. Study design and geographic distribution of included studies (N = 34)

Characteristic	n	%
Study design		
Quantitative (incl. 11 cross-sectional)	16	47.1%
Qualitative	11	32.4%
Mixed methods	7	20.6%
Region		
Europe	18	52.9%
North America	8	23.5%
Asia	5	14.7%
Africa	2	5.9%
South America	1	2.9%

Percentages are of all included studies (N = 34) and may not total 100% because of rounding. [Verify design classifications against your extraction table.]

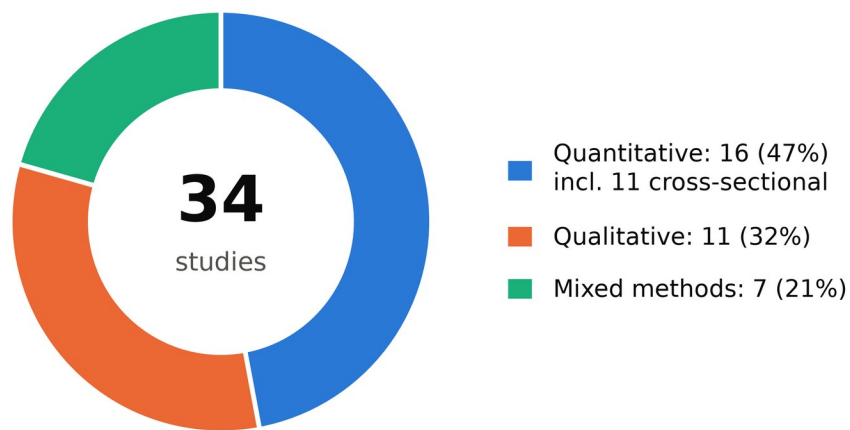


Figure 2. Distribution of included studies by design (N = 34). Cross-sectional studies (n = 11) are a subset of quantitative designs.

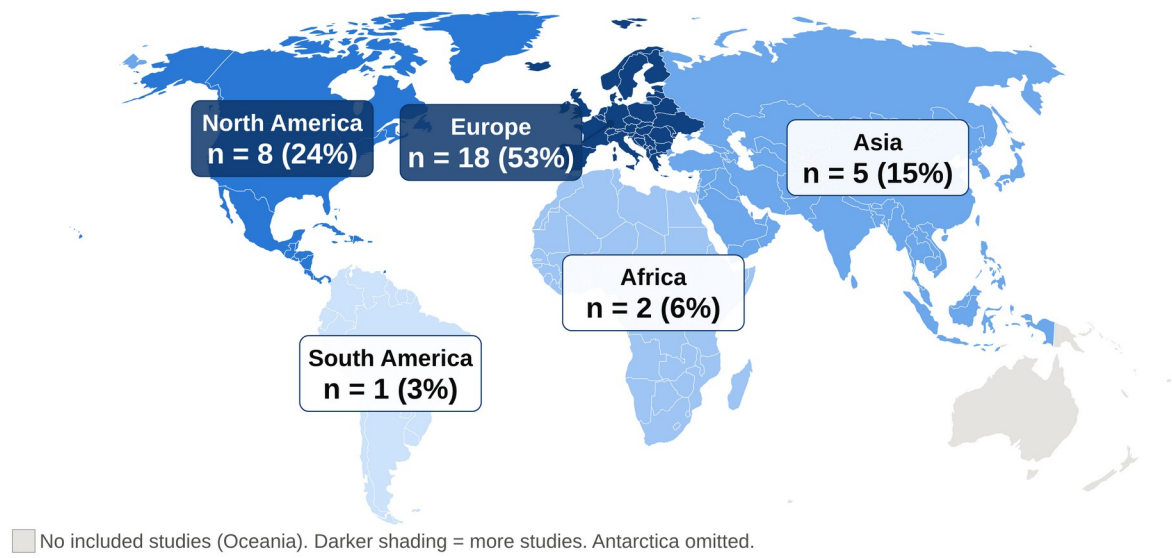


Figure 3. Geographic distribution of included studies by region (N = 34). Darker shading indicates more studies; no included study originated in Oceania.

Mental Health and Psychosocial Outcomes

Four outcome domains were identified (Table 2; Figure 4). Depression and anxiety were the most frequently reported outcomes, appearing in 26 of 34 studies (76.5%). Reduced quality of life and body image concerns were reported in 22 studies (64.7%), psychosexual and relational challenges in 19 (55.9%), and broader psychological distress in 17 (50.0%). These outcomes frequently co-occurred within the same studies, indicating that the psychosocial impact of MRKH extends across emotional, relational, and identity-related dimensions rather than being confined to a single domain.

Table 2. Mental health and psychosocial outcome domains reported in included studies (N = 34)

Outcome domain	Description	Studies, n (%)
Depression and anxiety	Depressive and/or anxiety symptoms or diagnoses	26 (76.5%)
Reduced quality of life and body image	Lower overall or mental health-related quality of life; negative body or genital image; self-esteem concerns	22 (64.7%)
Psychosexual and relational challenges	Sexual function and sexual self-esteem difficulties; concerns about intimacy, partner relationships, and disclosure	19 (55.9%)
Broader psychological distress	General distress, grief and loss, shame, isolation, and identity-related distress	17 (50.0%)

Studies could report more than one domain; counts are therefore not mutually exclusive. [Check that the domain descriptions match how studies were coded in your extraction.]

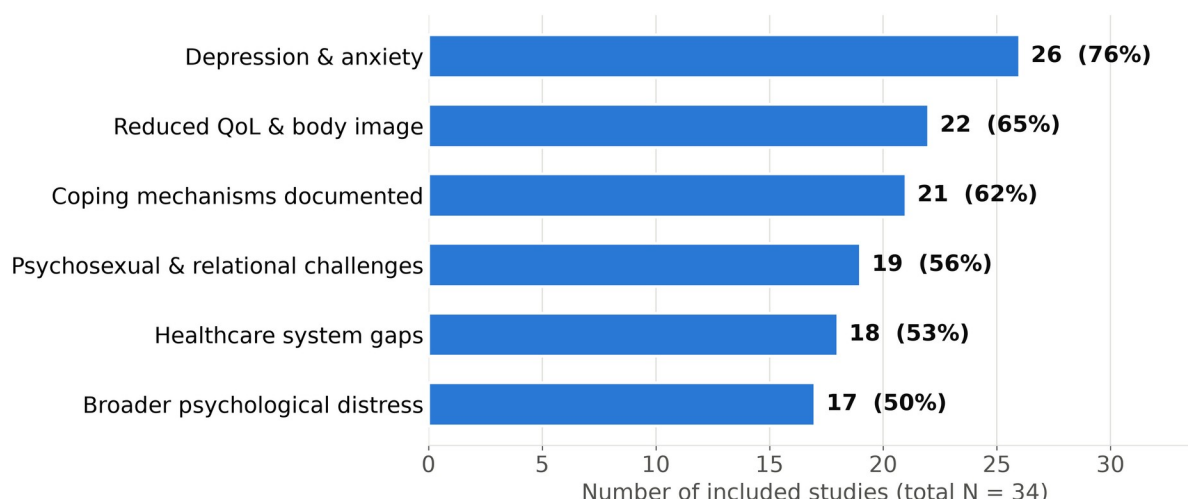


Figure 4. Number of included studies reporting each outcome domain, coping mechanisms, and healthcare system gaps (N = 34). Studies could contribute to more than one category.

Coping Mechanisms

Coping mechanisms were documented in 21 studies (61.8%). Six categories were identified (Table 3). Five were adaptive—peer and community support, psychological counseling, identity reconstruction, spiritual coping, and adaptive acceptance—and one, avoidance, was maladaptive. Consistent with the healthcare system gaps described below, these coping mechanisms were not supported by systematic clinical pathways.

Table 3. Coping mechanisms documented in included studies

Coping mechanism	Type	Description
Peer and community support	Adaptive	Connection with others with MRKH through support groups, online communities, and advocacy networks
Psychological counseling	Adaptive	Professional psychological support, including individual or group therapy
Identity reconstruction	Adaptive	Redefining femininity, womanhood, and self-concept independent of reproductive anatomy
Spiritual coping	Adaptive	Drawing on faith, religious practice, or spirituality to find meaning and resilience
Adaptive acceptance	Adaptive	Gradual acceptance of the diagnosis and its implications for fertility and sexuality
Avoidance	Maladaptive	Avoiding discussion of the diagnosis, concealment, and withdrawal from intimate relationships or care

[Confirm that the descriptions reflect the included studies; optionally add the number of studies reporting each coping mechanism from your extraction table.]

Healthcare System Gaps

Healthcare system gaps were identified in 18 studies (52.9%; Table 4). These included delayed diagnosis in adolescence without concurrent psychosocial support; limited access to multidisciplinary care teams that include psychology, social work, and gynecology; insufficient integration of mental health into MRKH care models; and the absence of standardized mental health screening protocols at diagnosis. In addition, the geographic distribution of the evidence base itself (Figure 3) reveals a research gap: only 3 of 34 studies originated in Africa or South America.

Table 4. Healthcare system gaps identified and corresponding public health implications

Gap identified	Public health implication
Delayed diagnosis in adolescence without concurrent psychosocial support	Integrate mental health screening into standard MRKH diagnostic pathways
Limited access to multidisciplinary care teams (psychology, social work, gynecology)	Develop multidisciplinary care protocols that include psychology and social work
Mental health insufficiently integrated into MRKH care models	Train providers in psychosocially informed MRKH care; build peer support infrastructure within reproductive health services
Lack of standardized mental health screening protocols at diagnosis	Adopt validated screening instruments at diagnosis and follow-up
Underrepresentation of low- and middle-income settings in the evidence base (research gap)	Expand research investment in underrepresented and low-resource settings

Discussion

Summary of Evidence

This scoping review mapped 34 studies published between 2019 and 2026 on the mental health and psychosocial outcomes of individuals with MRKH syndrome. Depression and anxiety were reported in about three of every four included studies, and reduced quality of life, body image concerns, psychosexual challenges, and broader psychological distress were each reported in at least half. Coping mechanisms were documented in about three in five studies, yet they

appeared to operate largely outside systematic clinical pathways, and more than half of the studies identified healthcare system gaps that limit access to psychosocial support.

These findings are consistent with earlier evidence. Before 2019, women with MRKH were shown to have greater psychological distress and lower self-esteem than controls,⁴ adolescents with MRKH had higher anxiety scores than healthy peers,⁵ and adult women reported poorer mental health–related quality of life and sexual wellness than normative samples.⁶ Prior systematic reviews similarly concluded that MRKH may be associated with psychological symptoms and impaired quality of life, particularly poor sexual esteem and genital image.^{9,10} The present review indicates that these concerns remain prominent in the recent literature and extends earlier syntheses by mapping coping mechanisms and healthcare system gaps alongside outcomes.

Coping and Psychosocial Support

The coping strategies identified—peer and community support, psychological counseling, identity reconstruction, spiritual coping, and acceptance—suggest that many individuals develop considerable resilience after diagnosis. Identity reconstruction aligns with qualitative accounts in which young women with MRKH worked to manage threats to identity and intimacy.⁸ At the same time, avoidance was documented as a maladaptive response that may delay help-seeking and treatment engagement. Evidence that structured psychological intervention can help is available: a randomized controlled trial of a cognitive-behavioural group intervention improved psychological outcomes among women with MRKH compared with a waiting-list control.¹³ The persistence of self-directed coping in the recent literature suggests that such evidence-based support has not been consistently translated into routine care.

Healthcare System Gaps and Equity

Although professional guidance identifies psychosocial counseling as central to MRKH management,¹ the gaps identified in this review—diagnosis without concurrent psychosocial support, limited multidisciplinary care, and no standardized mental health screening—indicate a gap between recommendations and practice. The concentration of evidence in Europe and North America (26 of 34 studies) further limits understanding of how MRKH is experienced where cultural expectations regarding fertility, marriage, and womanhood, as well as access to specialized care, may differ substantially. The scarcity of studies from Africa and South America represents both a research gap and a likely inequity in diagnosis, support, and long-term outcomes.

Implications for Practice, Policy, and Research

Five priorities emerge from this review. First, mental health screening should be integrated into standard MRKH diagnostic pathways, beginning at the time of diagnosis. Second, multidisciplinary care protocols should include psychology and social work alongside gynecology. Third, peer support infrastructure should be embedded within reproductive health services, building on the coping strategies individuals already use. Fourth, providers should be trained in psychosocially informed MRKH care, including sensitive communication about fertility and sexuality. Fifth, research investment should be expanded in low- and middle-income and underrepresented settings. Future studies should use standardized, validated mental health measures, longitudinal designs that follow individuals from diagnosis into adulthood, and trials of psychosocial interventions delivered within routine care.

Strengths and Limitations

Strengths of this review include the use of an established methodological framework, reporting according to PRISMA-ScR, a search of four major biomedical and psychological databases, and inclusion of quantitative, qualitative, and mixed-methods evidence. Several limitations should be noted. The review was limited to studies published from 2019 to 2026 and may therefore exclude earlier foundational research. Heterogeneity in study designs and outcome measures limits direct comparison across studies. Most studies came from high-income countries, limiting global generalizability, and some studies relied on self-reported mental health outcomes. Consistent with scoping review methodology, study quality was not formally appraised, and the counts reported here reflect how frequently outcomes were studied, not their prevalence. [Add, if applicable: English-language restriction; grey literature not searched.]

Conclusions

MRKH-related psychosocial burden is substantial yet under-integrated into care models. Depression, anxiety, reduced quality of life, and psychosexual challenges are frequently reported and often co-occurring, while coping mechanisms—though documented—remain unsupported by systematic clinical pathways. Multidisciplinary, mental health-inclusive care and greater research investment are urgently needed, particularly in underrepresented and low-resource settings, to address persistent inequities in diagnosis, support, and long-term outcomes.

Declarations

Acknowledgements: This scoping review was conducted under the mentorship of Dr. Paul Okojie and Dr. Robyn Anderson, Department of Public and Community Health, Liberty University.

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Conflicts of interest: [The authors declare no conflicts of interest. — confirm with all authors.]

Author contributions: [e.g., I.P.S.: conceptualization, search, screening, data charting, analysis, writing—original draft; P.O. and R.A.: supervision, methodology, writing—review and editing.

Edit to reflect actual contributions (CRediT taxonomy).]

Ethics approval: Not applicable; this review analyzed published studies.

Data availability: The data charting form and extracted data are available from the corresponding author on reasonable request.

Use of AI tools: [Disclose any AI-assisted writing or editing per the target journal's policy.]

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[Add citations for the 34 included studies (numbered in order of first citation, e.g., in Appendix B) per the target journal's style.]

Appendix A. Search Strategy

[Paste the exact search string run in each database, the platform/interface used, the limits applied, the date each search was run, and the number of records retrieved per database (these should total 516).]

Table A1. Records retrieved by database

Database	Platform	Date searched	Records retrieved
PubMed/MEDLINE			
Scopus			
PsycINFO			
CINAHL			
Total			516

Appendix B. Characteristics of Included Studies

[Complete one row per included study from your data-charting form. Column totals must match the counts reported in the Results.]

Table B1. Characteristics of included studies (N = 34)

#	Author (year)	Country	Design	Sample (n; age)	Mental health outcomes	Coping mechanisms	Healthcare gaps
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Appendix C. PRISMA-ScR Checklist

Table C1. PRISMA-ScR checklist (Tricco et al., 2018)

Section / topic	Item	Reported in
Title	1	Title page
Structured summary	2	Abstract
Rationale	3	Introduction
Objectives	4	Introduction (research question)
Protocol and registration	5	Methods: Study Design [complete]
Eligibility criteria	6	Methods: Eligibility Criteria
Information sources	7	Methods: Information Sources; Appendix A
Search	8	Appendix A [complete]
Selection of sources of evidence	9	Methods: Study Selection
Data charting process	10	Methods: Data Charting
Data items	11	Methods: Data Charting
Critical appraisal of individual sources of evidence (optional)	12	Methods: Synthesis (not performed)
Synthesis of results	13	Methods: Synthesis
Selection of sources of evidence	14	Results; Figure 1 [add full-text stage]
Characteristics of sources of evidence	15	Results; Table 1; Appendix B
Critical appraisal within sources of evidence (optional)	16	Not applicable
Results of individual sources of evidence	17	Appendix B
Synthesis of results	18	Results; Tables 2–4; Figures 2–4
Summary of evidence	19	Discussion: Summary of Evidence
Limitations	20	Discussion: Strengths and Limitations
Conclusions	21	Discussion: Conclusions
Funding	22	Declarations